AMOD5430/COIS3510 Data Visualization – Workshop 2 Instructions

Configure Your Environment

Please follow these instructions to set up your computer environment:

- Install R and RStudio on Windows
- Install R and RStudio on Mac
- Install R and RStudio on Ubuntu

Alternatively, you may set up a Python-based environment:

- Installing Python on Windows
- Installing Python on Mac
- Installing Matplotlib on Windows
- Installing Matplotlib
- Installing Matplotlib on Mac
- Installing Jupyter on Windows
- Installing Jupyter on Mac

Displaying Multivariate Data in R

Open RStudio and complete the following plots by entering code snippets below. Please submit a PDF file containing the charts you generated, and the rmd file, to the Blackboard assignment.

The final PDF MUST include the original plot as well as the exercise plot

Basic Scatter Plot

```
# Basic scatter plot
x <- c(6,7,8,7,2,3,9,4,10,15,9,7)
y <- c(98,85,87,88,112,104,87,94,78,78,85,86)
plot(x,y)
```

Exercise: Add the x-axis label "Hours Studied" to the scatter plot.

Scatter Plot with Multiple Classes

```
# scatter plot with multiple classes
plot(iris$Petal.Length, iris$Petal.Width, pch=c(23,24,25)[unclass(iris$Species)], main="Iris Data")
```

Exercise: Change the colour palette so that each iris species is represented by a different colour.

Scatter Plot with 4 Dimensional Data

```
# Scatter plot with 4 dimensions (petal lenght added)
attach(iris)
plot = plot(Sepal.Length,Sepal.Width,type="p",
            cex = ,
            bg = Species, # Background color

        pch = 21)
```

Exercise: Map petal length to the size of the plotting points. (use cex =" " which is character expansion)

Scatter Plot Matrix

```
# scatter plot matrix
pairs(~Sepal.Length+Sepal.Width+Petal.Length,data=iris,
      upper.panel = NULL)
```

Exercise: Add Petal Width as an element to the scatter plot matrix.

Scatter Plot Lattice

```
# scatter plot lattice
library("lattice")
xyplot(Sepal.Length ~ Petal.Length,
        group = Species,
        data = iris,
        type = c("p", "smooth"),
        scales = "free")
```

Exercise: Create a separate plot for each species

Parallel Axes

```
# parallel axes
library(GGally)
ggparcoord(data,
           columns = 1:4)
```

Exercise: Change the plot so that the lines are coloured by iris species.

Chernoff Faces

```
# Chernoff Faces
install.packages("aplpack")
```

```
library("aplpack")

# View the faces

faces(face.type=1)

# View the effect of the variables

faces(rbind(1:3,5:3,3:5,5:7))


# View the data

data(longley)
faces(longley[1:9,],face.type=0)
```

Exercise: Generate Chernoff faces using the first six observations of the Longley dataset.

```
# Using in a scatter plot

plot(longley[1:16,2:3],bty="n")
a<-faces(longley[1:16,],plot=FALSE)
plot.faces(a,longley[1:16,2],longley[1:16,3],width=35,height=30)
```

Radar Charts

```
# Basic chart

install.packages("fmsb")
library("fmsb")
df <- data.frame(rbind(rep(10, 8), rep(0, 8),
                       matrix(sample(0:10, 8),
                              nrow = 1)))


colnames(df) <- paste("Var", 1:8)
radarchart(df)
```

Exercise: Change the colour of the radar chart's outline to red.

```
# Customize

radarchart(df,
           cglty = 1,       # Grid line type
           cglcol = "gray", # Grid line color
           cglwd = 1,       # Line width of the grid
           pcol = 4,        # Color of the line
           plwd = 2,        # Width of the line
           plty = 1)        # Line type of the line


# Multiple groups or classes

set.seed(1)
df2 <- data.frame(rbind(rep(10, 8), rep(0, 8),
```

```
                        matrix(sample(0:10, 24,
                                      replace = TRUE),
                               nrow = 3)))

colnames(df2) <- paste("Var", 1:8)
radarchart(df2,
           cglty = 1,       # Grid line type
           cglcol = "gray", # Grid line color
           pcol = 2:4,      # Color for each line
           plwd = 2,        # Width for each line
           plty = 1)        # Line type for each line


# Fill the areas

areas <- c(rgb(1, 0, 0, 0.25),
           rgb(0, 1, 0, 0.25),
           rgb(0, 0, 1, 0.25))
radarchart(df2,
           cglty = 1,       # Grid line type
           cglcol = "gray", # Grid line color
           pcol = 2:4,      # Color for each line
           plwd = 2,        # Width for each line
           plty = 1,        # Line type for each line
           pfcol = areas)   # Color of the areas
```

Displaying Multivariate Data in Python/Jupyter

Start your Jupyter Lab and complete the following plots by entering code snippets below. Please submit a PDF file containing the charts you generated, and the Jupyter notebook source file, to the Blackboard assignment.

```
import matplotlib.pyplot as plt

import matplotlib.cm

import seaborn as sns

import pandas as pd

from pandas.plotting import parallel_coordinates

from ChernoffFace import *

import numpy as np
```

Basic Scatter Plot

```
# Define the data
x = [6, 7, 8, 7, 2, 3, 9, 4, 10, 15, 9, 7]
y = [98, 85, 87, 88, 112, 104, 87, 94, 78, 78, 85, 86]

# Create the scatter plot
plt.scatter(x, y)

# Display the plot
plt.show()
```

Exercise: Add the x-axis label "Hours Studied" to the scatter plot.

Scatter Plot with Multiple Classes

```
# Load the iris dataset
iris = sns.load_dataset('iris')

# Create a scatter plot with different species as different classes
sns.scatterplot(x='petal_length', y='petal_width', style='species', data=iris)

# Add title
plt.title('Iris Data')
plt.show()
```

Exercise: Change the colour palette so that each iris species is represented by a different colour.

Scatter Plot with 4 Dimensional Data

```
# Create a scatter plot with Sepal.Length vs Sepal.Width
```

```
sns.scatterplot(x='sepal_length', y='sepal_width', hue='species', data=iris, sizes=(20, 200), style='species', palette='viridis')
```

Exercise: Map petal length to the size of the plotting points.

```
# Add title and show plot
plt.title('Iris Data - Sepal Dimensions with Petal Length as Size')
plt.show()
```

Scatter plot Matrix

```
# Create a scatter plot matrix
sns.pairplot(iris, diag_kind='hist')

# Show the plot
plt.show()
```

Exercise: Create a scatter plot matrix using only Sepal Length, Sepal Width, Petal Length and Petal Width.

Scatter Plot Lattice

```
# Create a faceted scatter plot lattice with a smoothing line
g = sns.lmplot(x='petal_length', y='sepal_length', col=, hue='species', data=iris,
        height=4, aspect=0.7, scatter_kws={'s': 50}, line_kws={'color': 'black'}, ci=None)
```

Exercise: Create a separate plot for each species

```
# Customize the scales and titles
g.set_axis_labels('Petal Length', 'Sepal Length')
g.set_titles('Species: {col_name}')
```

```
# Show the plot
plt.show()

##### Parallel Axes

# Load the iris dataset
url = "https://raw.githubusercontent.com/vincentarelbundock/Rdatasets/master/csv/datasets/iris.csv"
iris = pd.read_csv(url)

# Parallel coordinates plot
plt.figure(figsize=(10, 6))
parallel_coordinates(iris, 'Species', cols=['Sepal.Length', 'Sepal.Width', 'Petal.Length', 'Petal.Width'],
color=['r', 'g', 'b'])
```

Exercise: Change the plot to display only Sepal Length and Petal Length

```
# Add title and show the plot
plt.title('Parallel Coordinates Plot - Iris Data')
plt.xlabel('Features')
plt.ylabel('Values')
plt.show()

##### Chernoff Faces

##### Things to run

### - pip install ChernoffFace

### - ModuleNotFoundError: No module named 'pkg_resources' --> pip install setuptools

# Load the Longley dataset from the URL
url = "https://github.com/vincentarelbundock/Rdatasets/raw/master/csv/datasets/longley.csv"
data = pd.read_csv(url)
```

```
# Generate Chernoff faces with the Longley data
fig = chernoff_face(data=data,
                    titles=[str(x) for x in list(range(len(data)))],
                    color_mapper=plt.cm.Pastel1)
```

Exercise: Generate Chernoff faces using the first six observations of the Longley dataset.

```
# Display
fig.tight_layout()
plt.show()


##### Radar Charts


# Data
labels = ['Var 1', 'Var 2', 'Var 3', 'Var 4', 'Var 5', 'Var 6', 'Var 7', 'Var 8']
num_vars = len(labels)

# Create the data for the radar chart
data = [np.random.randint(0, 11) for _ in range(num_vars)]
data += data[:1]  # Close the loop by repeating the first value

# Define the angle for each axis (in radians)
angles = np.linspace(0, 2 * np.pi, num_vars, endpoint=False).tolist()
angles += angles[:1]  # Complete the loop by appending the first angle

# Customize the chart (similar to R's grid line type, colors, and line width)
fig, ax = plt.subplots(figsize=(6, 6), subplot_kw=dict(polar=True))

# Grid lines
ax.spines['polar'].set_visible(True)
```

```
ax.grid(color='gray', linestyle='-', linewidth=1)

# Fill the area under the radar chart
ax.fill(angles, data, color='blue', alpha=0.25)

# Plot the outline of the radar chart
ax.plot(angles, data, color='blue', linewidth=2, linestyle='-')
```

Exercise: Change the colour of the radar chart's outline to red.

```
# Remove the radial labels (y-axis labels)
ax.set_yticklabels([])

# Set the ticks (labels around the radar chart)
ax.set_xticks(angles[:-1])
ax.set_xticklabels(labels)
plt.show()
```